

Principles of communication systems

EET3202, CUNY City Tech, Fall 2023

Homework #02 Solutions

S. M. Farzaneh

Problem 1

Part 1

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$

$$g(t) \xleftrightarrow{\mathcal{F}} G(f)$$

Change of variables from f to $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi(-f)t} dt = \int_{-\infty}^{\infty} g(t) e^{j2\pi ft} dt$$

change of variables from t to $-t$

$$G(-f) = \int_{-\infty}^{\infty} g(-t) e^{-j2\pi ft} dt, \text{ Therefore } g(-t) \xleftrightarrow{\mathcal{F}} G(-f)$$

Problem 1

Part 2

$$\begin{aligned} g(t) &= e^{-at} u(t), & G(f) &= \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt \\ G(f) &= \int_{-\infty}^{\infty} u(t) e^{-(j2\pi f + a)t} dt = \int_0^{\infty} e^{-(j2\pi f + a)t} dt \\ &= \frac{-1}{a + j2\pi f} e^{-(a + j2\pi f)t} \Big|_0^{\infty} = 0 - \left(\frac{-1}{a + j2\pi f} \right) = \frac{1}{a + j2\pi f} \end{aligned}$$

Problem 1

Part 3

$$g(t) = e^{-at} u(t)$$

$$g(t) \xleftrightarrow{\mathcal{F}} G(f) = \frac{1}{a + j2\pi f}$$

$$g(-t) = e^{at} u(-t)$$

$$g(-t) \xleftrightarrow{\mathcal{F}} G(-f) = \frac{1}{a - j2\pi f}$$

$$v(t) = e^{-a|t|} = \begin{cases} e^{-at} & t \geq 0 \\ e^{at} & t < 0 \end{cases}$$

$$= e^{-at} u(t) + e^{at} u(-t) = g(t) + g(-t)$$

$$v(t) \xleftrightarrow{\mathcal{F}} V(f) = G(f) + G(-f) = \frac{1}{a + j2\pi f} + \frac{1}{a - j2\pi f} = \frac{2a}{a^2 + (2\pi f)^2}$$

Problem 2

$$\text{sgn}(t) = \begin{cases} 1 & , t > 0 \\ 0 & , t = 0 \\ -1 & , t < 0 \end{cases} = \lim_{a \rightarrow 0} f_a(t) = \lim_{a \rightarrow 0} \left(e^{-at} u(t) - e^{at} u(-t) \right)$$

$$\mathcal{F}\{\text{sgn}(t)\} = \mathcal{F}\left\{\lim_{a \rightarrow 0} f_a(t)\right\} = \lim_{a \rightarrow 0} \mathcal{F}\{f_a(t)\}$$

from problem 1 we know

$$e^{-at} u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a + j2\pi f} \quad \text{and} \quad e^{at} u(-t) \xleftrightarrow{\mathcal{F}} \frac{1}{a - j2\pi f}$$

Therefore,

$$\mathcal{F}\{f_a(t)\} = \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f} = \frac{-j4\pi f}{a^2 + (2\pi f)^2}, \quad \lim_{a \rightarrow 0} \mathcal{F}\{f_a(t)\} = \frac{-j4\pi f}{4\pi^2 f^2} = \frac{1}{j\pi f}$$

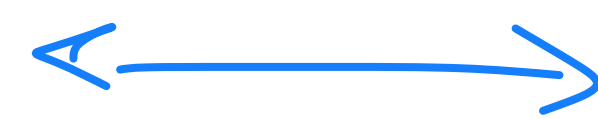
Problem 3

$$g(t) = e^{-a|t-t_0|}$$

Time Shifting property of Fourier Transform:

$$v(t) \xleftrightarrow{\mathcal{F}} V(f), \quad v(t-t_0) \xleftrightarrow{\mathcal{F}} V(f) e^{-j2\pi f t_0}$$

Shifting the signal in time by t_0



Multiplying Fourier transform by $e^{-j2\pi f t_0}$

from problem 1:

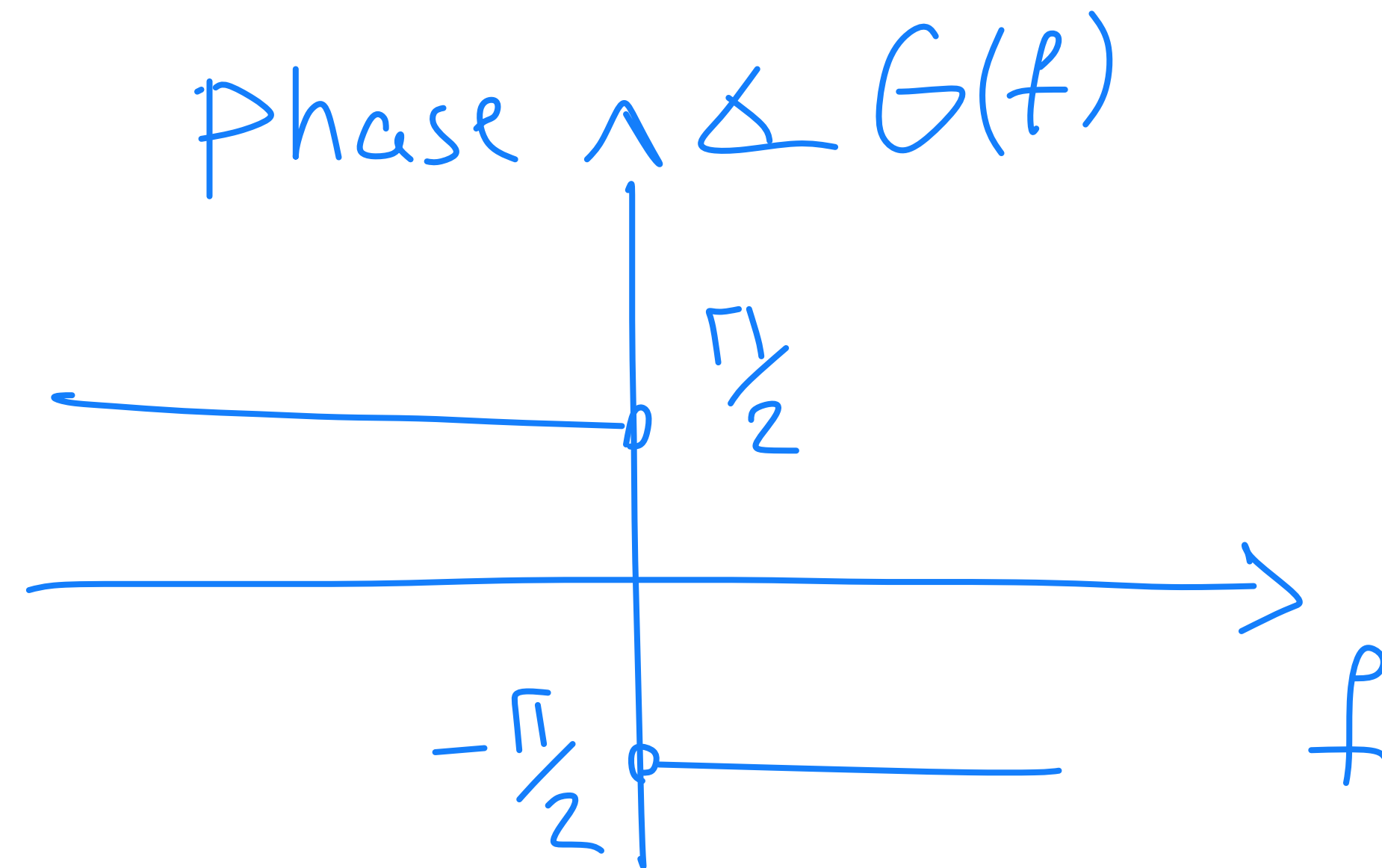
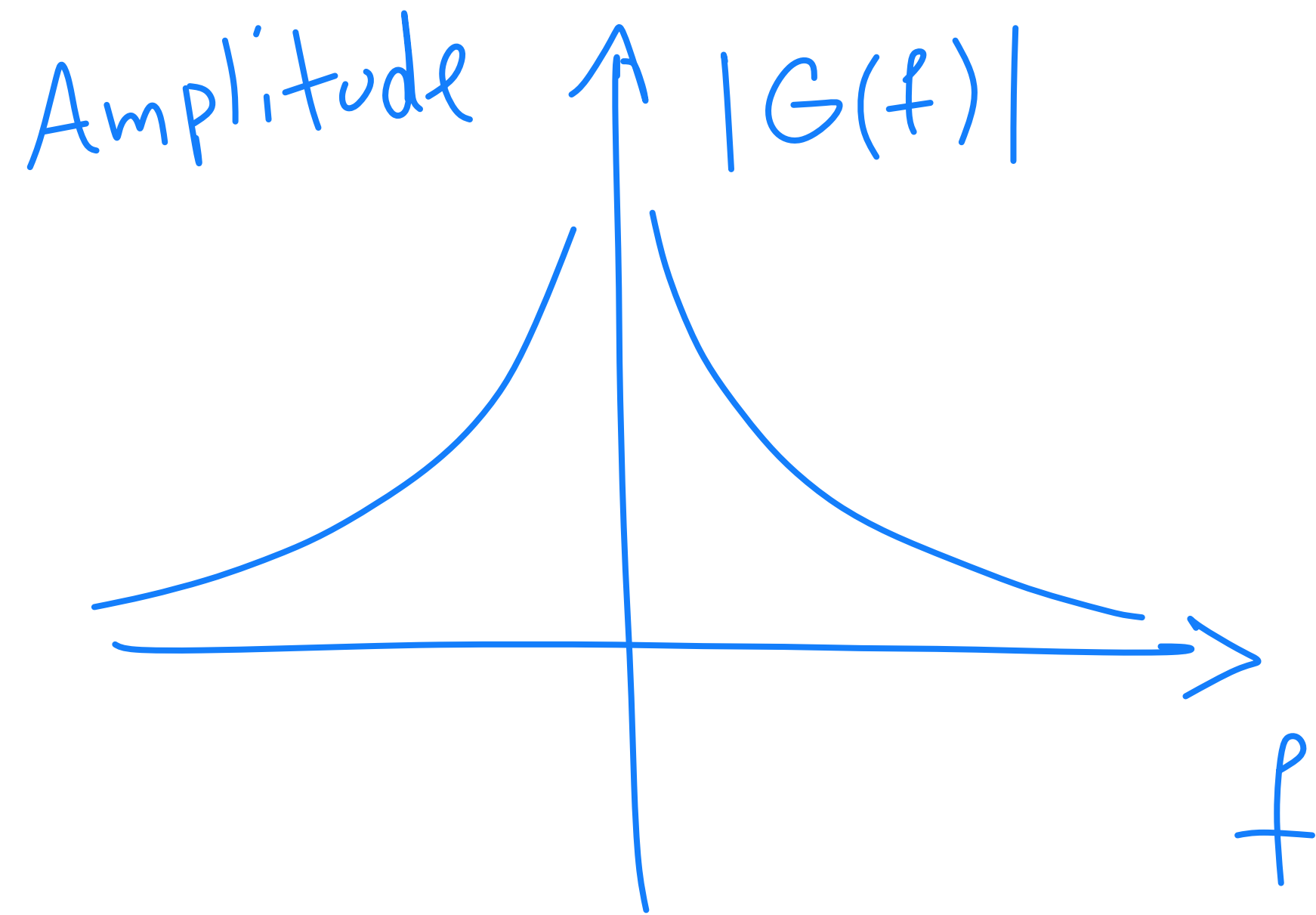
$$e^{-a|t|} \xleftrightarrow{\mathcal{F}} \frac{2a}{a^2 + (2\pi f)^2}$$

Therefore,

$$e^{-a|t-t_0|} \xleftrightarrow{\mathcal{F}} \frac{2a}{a^2 + (2\pi f)^2} \cdot e^{-j2\pi f t_0}$$

Problem 4

$$G(f) = \frac{1}{j2\pi f} = \frac{1}{2\pi f}(-j) = \frac{1}{2\pi f} e^{j(-\frac{\pi}{2})} = \begin{cases} \left| \frac{1}{2\pi f} \right| e^{j(-\frac{\pi}{2})} & f > 0 \\ \left| \frac{1}{2\pi f} \right| e^{j(-\frac{\pi}{2} + \pi)} & f < 0 \end{cases}$$



* Amplitude should be always positive